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Module 1: Introduction
Data and Notation
Parameters

Lesson 2: Conjugate Priors
and Multivariate Normal
Models

Reading: Conjugate Priors
and Multivariate Normal
Models Readings
1h

Video: Conjugate Priors Pt. 1
7 min

Video: Conjugate Priors Pt. 2
8 min

Video: Conjugate Priors Pt. 3
7 min

Practical Assignment:
Conjugate Priors and
Multivariate Normal
Models Quiz
Submitted

Lab: Module 3 - Lesson 2 -
RStudio Lab
1h

Lesson 3: Advanced
Multivariate Models and
Practical Applications

Module 3 Summative
Assessment

Module 3 Summary

Your grade: 100%

Your best: 100% - Your highest score

Next item →

Conjugate Priors and Multivariate Normal Models Quiz

Instructions

Review Learning Objectives

1. What is the scaled inverse- χ^2 distribution used for in Bayesian analysis of normal data? 1 / 1 point

- ☐ To model the posterior distribution of the mean
- ☐ To model the mean of the data
- ☒ To model the variance of the data
- ☐ To model the joint prior distribution

Correct

Assignment details
The scaled inverse- χ^2 distribution is used to model the variance of the data in the Bayesian analysis of normal data.

Submitted

Attempts

Unlimited

Retry

2. How should you sample from the conjugate posterior distribution for data normally distributed with unknown mean and unknown variance? 1 / 1 point

100%
To get you need at least 80%. We keep your highest score.

View submission

See feedback

- ☒ Sample first the variance then the mean
- ☐ Take both at the same time
- ☐ Sample the mean first and then the variance
- ☐ It is not possible to sample these parameters from the posterior

Correct

Correct! It is easiest to sample the variance from the inverse chi squared distribution based on the data and then sample the mean from the marginal posterior distribution.

Like

Dislike

Report an issue

3. Suppose a study aimed to assess the preferred mode of transportation among three options: Car, Bicycle, and Bus. Out of 1200 participants:

- 400 preferred Car
- 300 preferred Bicycle
- 300 preferred Bus

Compute the 95% credibility interval for those who prefer to travel by Bus.

- ☒ 0.21 to 0.29
- ☐ 0.13 to 0.45
- ☐ 0.01 to 0.11
- ☐ 0.0 to 1.0

Correct

Correct! Your simulations should produce something roughly in this range.